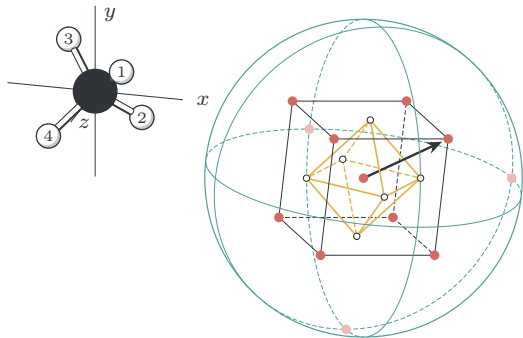
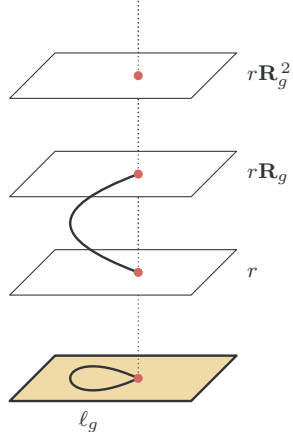


Methane, CH_4 : the reference $\mathbf{X}_0$ in its body frame,
with the C_2 axes x, y, z through the midpoints of opposite edges.



The orientation space $SO(3)$ as a ball:
direction the axis, distance the angle.
Red dots are the twelve rotations of T ;
the gold octahedron is the cell $\mathcal{U}$ at $q = 0$;
the arrow is the lift $r \rightarrow r\mathbf{R}_g$ of the
third-turn about bond 1.



Over the base $SO(3)/T$ lie twelve sheets,
one for each rotation in T ; three are drawn.
The third-turn is a closed loop ℓ_g in the base;
lifted to the sheets it does not close,
but climbs from sheet r to sheet $r\mathbf{R}_g$.
A species sees the climb as a number or matrix:
A as 1, ${}^1\text{E}$ as ω , ${}^2\text{E}$ as ω^2 , T as $x \rightarrow y \rightarrow z$.